1.4 Perimeter and Area in the Coordinate Plane

Name _____ Period ____ Date _____

Geometry 1.4 (30-minute plan) · Student sheet

I can…

- classify a polygon by its number of sides and as convex or concave.
- find the perimeter of a polygon in the coordinate plane.
- find the area of a polygon in the coordinate plane.

Vocabulary · fill in the missing word

Polygon — a closed plane figure formed by three or more segments that meet only at their _____

n-gon — a _____ with n sides

Convex polygon — a polygon in which no line that _____ a side passes through the interior

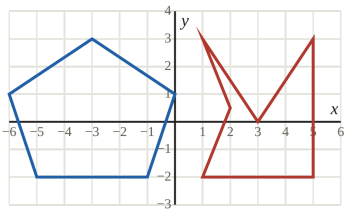
Concave polygon — a polygon that is not _____

Part 1 · Notes and examples (fill in each note, then work the example)

Classifying polygons: name a polygon by its number of sides; it is convex if no line containing a side passes through the _____.

Example 1

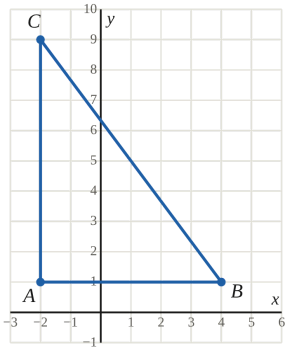
Classify each polygon by its number of sides. Tell whether it is convex or concave.



Perimeter on the coordinate plane: count horizontal and vertical sides; use the _____ for each slanted side; then add.

Example 2

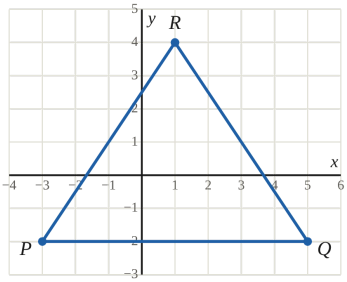
Find the perimeter of △ABC with A(−2, 1), B(4, 1), and C(−2, 9).



Area: for a triangle, A = ½bh, where the height is _____ to the base.

Example 3

Find the area of $\triangle PQR$ with P(−3, −2), Q(5, −2), and R(1, 4).



Part 2 · Partner practice

Partner practice

Find the area of the rectangle with vertices (−2, −1), (4, −1), (4, 3), and (−2, 3).